Reliability Engineering and System Safety 203 (2020) 107048

Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

General multi-state rework network and reliability algorithm

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ARTICLE INFO

Keywords:
Analytical results
Discrete-event simulation
System reliability
Rework

ABSTRACT

A rework network is a common manufacturing system, in which flows (products) are processed in a sequence of workstations (nodes), which often results in defective products. To improve the productivity and utility of the system, the rework network allows some of the defective products to go back to the "as normal" condition after the rework process. Since 2017 [1, 2], the traditional study proposed an algorithm to correct more than 21 archive publications regarding the rework network reliability problem, which is an important real-life problem. In addition to traditional research, we provide an accurate general model based on the novel state distribution with a smaller number of limitations. Furthermore, we propose an algorithm to calculate the reliability of the multi-state rework networks using the proposed novel state distributions.

1. Introduction

The rework network is a special network with a rework process, which is popular in reworking defective products back to the "as normal" condition [1–7]. A traditional rework network can be considered to be a simple network but with both the deterioration effect [8–10] and the learning effect [11–13]. During the transmission and processing of the flow (products) from one node (such as workstation, machine, or worker) to another in the rework network, the amount of flow decreases owing to the deterioration effect of the discarded/defective products [8–10]. However, in some nodes, the amount of flow increases owing to the learning effect [11–13], resulting from some detective products being able to be processed again after the rework to fix/repair them.

A rework network $G(V, E, \mathbf{D})$ comprises a node set $V = \{1, 2, \ldots, n\}$ and an arc set $E = \{e_{i,j} \mid i, j \in V\}$. $\mathbf{D}$ is the state distribution defining the state levels $\{0, 1, \ldots, \mathbf{D}_{\max}(k)\}$ and the occurrent probability $\mathbf{D}(k, l) = p_{k,l}$ of the related state level $l \in \{0, 1, \ldots, \mathbf{D}_{\max}(k)\}$ and the defect rate $\mathbf{D}(k) = p_k$ for all nodes $k \in V$. In $G(V, E, \mathbf{D})$, each node could be either a process station, work station, unit, machine, tool, or worker to process (including any type of manufacture) the products that are input to the rework network, and the flow could be any product, such as oil or a car [14–16].

Let node β be a predecessor of node α, where $\alpha \neq \beta$ and $\alpha, \beta \in V$. For simplicity, nodes α and β are referred to as the rework-input node and the rework-begin node to represent the node that sends defects to the rework process and the first node in the rework process, respectively. Hence, node α sends the defective products back to node β to be reworked in the rework process.

Note that [1,2]:

1 Except for node α, all the other nodes discard the defective products immediately after they are identified.
2 Each defect has only one chance to be reworked, i.e., it cannot be reworked again if it fails after it has been sent back from node α.

For example, the network illustrated in Fig. 1 is a rework network $G(V, E, \mathbf{D})$, where $V = \{1, 2, 3, 4\}$ and $E = \{e_{1,2}, e_{2,3}, e_{3,4}\}$ [1,2]. The state distribution $\mathbf{D}$ is listed in Table 1. Node 0, indicated by a dotted circle is a pseudo node (i.e., not real), which is used to express the values of input flows.

Among these four nodes, there is one special arc from node 3 to node 2 in Fig. 1 to represent the rework process. Node 3 is the rework_input node α and node 2 is the rework_begin node β. Defects are only allowed to be sent back from node 3 to node 2 to be reworked, and the other nodes simply discard the defective products. For example, any

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https://doi.org/10.1016/j.ress.2020.107048
Received 23 June 2019; Received in revised form 4 May 2020; Accepted 22 May 2020
Available online 27 May 2020
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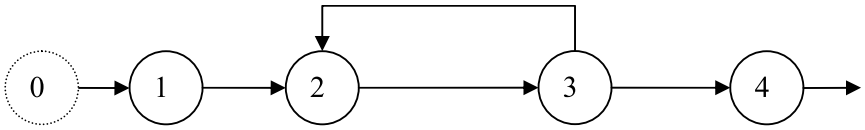


Fig. 1. Example rework network.

Table 1
State distribution **D** of Fig. 1.

k	$\mathbf{D}(k)=p_k$	State l	$\mathbf{D}(k,\ l)=p_{k,l}$	k	$\mathbf{D}(k)=p_k$	State l	$\mathbf{D}(k,\ l)=p_{k,l}$
1	0.9			3	0.9		
		0	0.01			0	0.005
		5	0.01			3	0.005
		10	0.05			6	0.010
		15	0.93			9	0.010
						12	0.010
						15	0.960
2	0.9			4	0.9		
		0	0.01			0	0.005
		5	0.02			3	0.005
		10	0.02			6	0.005
		15	0.95			9	0.010
						12	0.015
						15	0.960

defective product must be discarded if it is detected right after being processed in nodes 1, 2, or 4; however, it can be sent back to node 2 to be reworked if it is detected in node 3.

A path in a rework network is a procession to describe the way to process products in a rework network. For example, there are only two paths in Fig. 1: the normal path $p_n = \{1, 2, 3, 4\}$, which represents the process of perfect products without any defect, and the rework path $p_r = \{1, 2, 3, 2, 3, 4\}$, which represents the process of defective products by the rework process [1,2].

In a recent study, it pointed out that there exists an error in all the current publications related to the rework problem in network reliability, and proposed a rework network model, where in the number of defective products generated in each node followed the binomial distribution [1,2]. For example, in node i, the probability of having p_i and q_i units of perfect and defective products is $C_{q_i}^{p_i+q_i}(1-\delta_i)^{p_i}\delta_i^{q_i}$, where δ_i is the defect rate of node i. it also proposed a method for computing the multi-state two-terminal reliability in such a network through rework operations [1].

It is claimed that the proposed algorithm has the following advantages: (1) The algorithm's accuracy has been verified by using a discrete-event simulation, and a mathematical analysis of the errors published in previous papers has been provided. (2) The algorithm is a useful tool for assessing the validity of any future proposed approach for other stochastic reliability problems.

There are indeed some errors published in the more than 25 previous related papers cited in the traditional research paper that have been corrected [1]. As there are several important applications of the rework network in real-life as demonstrated in [1,2] and all related references provided in [1,2], there is a need to provide a more precise concept of the binomial rework network and we explore a general rework network with a more efficient algorithm to calculate its reliability.

A more general rework network is proposed by removing some impractical restrictions from the traditional rework network together with a novel algorithm for solving the reliability of the proposed generalized binomial rework network. The remainder of this paper is structured as follows. Section 2 presents the required notations of the proposed general rework network. The proposed general rework network is discussed in Section 3 together with its assumption, innovations, and a novel algorithm for its reliability. The proposed algorithm to calculate the reliability of the proposed general rework network is discussed in Section 4. Section 5 presents a performance analysis of the proposed algorithm. Finally, the conclusion is presented in Section 6.

2. Notations and assumptions

The required notations for the proposed general rework network and reliability algorithm are described in this section.

V	:	node set $V = \{1, 2, \ldots, n\}$
E	:	arc set $E = \{e_{i,j} \mid i, j \in V\}$
δ_k	:	defect rate of node $k \in V$ if not in the rework process
δ	:	rate of sending defects in the rework process
γ_k	:	defect rate of node $k \in V$ in the rework process
$\mathbf{D}_{\max}(k)$	:	maximal state level of node $k \in V$.
$\mathbf{D}_k$	:	state levels $\mathbf{D}_k = \{0, 1, \ldots, \mathbf{D}_{\max}(k)\}$
$\mathbf{D}(k,\ l)$	:	occurrent probability of the related state level $l \in \mathbf{D}_k$.
$\mathbf{D}$	:	state distribution defining $\mathbf{D}_{\max}(k)$, $\mathbf{D}(k,\ l)$, $\mathbf{D}_k$, δ_k, δ and γ_k for all $k \in V$.
$c_i(l)$	:	smallest state level of node i, which is not less than l, e.g., $c_2(2) = 5$ in Table 1.
$C_i(l)$	:	probability of having $c_i(l)$, e.g., $C_2(2) = 0.02$ in Table 1.
α	:	rework-input node that sends the defective products to the rework process, e.g., node 3 in Fig. 1
β	:	rework-begin node, which is the first node in the rework process, e.g., node 2 in Fig. 1
$G(V, E, \mathbf{D})$	:	rework (activity-on-node) network with $V = \{1, 2, \ldots, n\}$, $E = \{e_{i,j} \mid i, j \in V\}$, $\mathbf{D}$, $\alpha \in V$, and $\beta \in V$
P(●)	:	probability of the occurrence of the event ●
b	:	input value
b^*	:	input value to the rework node
d	:	output demand
R_d	:	system reliability that at least d units of output do not have any defect
F_o	:	total output
p_n	:	normal path, which is a path without rework
p_r	:	rework path
$F_o(p)$	:	output of the path p
$F_o(p,\ k)$	:	output of node k for path p
$F_i(p,\ k)$	:	input of node k for path p. Note that $F^*(p_n,\ k) = \min(F(p_n,\ j), \mathbf{D}_{\max}(k))$ if node j is the node right before node k in p_n.
p_x, $p_{x,i}$	:	non-rework vector $p_x = (p_{x,0}, p_{x,1}, p_{x,2}, \ldots, p_{x,n})$, which is the state vector without rework and $p_{x,i}$ is the number of perfect products in node i without using the rework process for $i = 1, 2, \ldots, n$, where $p_{x,0} = b$ and $p_{x,i} \le p_{x,j}$ if $j \le i$.
q_x, $q_{x,i}$	:	$q_y = (q_{x,0}, q_{x,1}, q_{x,2}, \ldots, q_{x,n})$ and $q_{x,i} = b - \sum_{k=1}^{i} p_{x,k}$ if p_x is the related non-rework vector
π_y, $\pi_{y,j}$	:	rework vector $\pi_y = (\pi_{y,\alpha}, \pi_{y,\beta}, \pi_{y,\beta+1}, \ldots, \pi_{y,n})$, which is the state vector right after rework and $\pi_{y,j}$ is the number of perfect products produced in node j from the rework process for $j = \beta, \beta+1, \ldots, n$, where $\pi_{y,\alpha} = b^*$ and $\pi_{y,i} \le \pi_{y,j}$ if $j \le i$
θ_y, $\theta_{y,j}$	:	$\theta_y = (\theta_{y,\alpha}, \theta_{y,\beta}, \theta_{y,\beta+1}, \ldots, \theta_{y,n})$ and $\theta_{y,i} = b^* - \sum_{k=\beta}^{i} \pi_{y,k}$ if π_y is the related rework vector
C_q^Q	:	$\frac{Q!}{q!(Q-q)!}$

3. Proposed generalized binomial rework network

The proposed rework network generalized from the traditional rework network is discussed in this section. We first address the limitation of the current rework network, and then generalize the rework network by considering more practical situations and providing more robust assumptions. Moreover, we provide solutions based on the multi-vector to denote the situations where the rework network is able to produce the required number of products to help in developing algorithms and calculating the rework network reliability.

3.1. Innovation and assumptions of the proposed rework network

In addition to the current rework network, the following assumptions are provided:

Assumption 1. The rework process is unnecessary for the defect products if and only if the number of all perfect products are greater than or equal to d. In the current rework network, there is no clear answer whether a rework is still needed if the number of final non-defective products are already met.

Assumption 2. If the rework process is required, it is implemented only after the normal processes of all the products are completed. It is unclear as to how the normal products and reworked products share the capacities of each node, as discussed in Section 2.4 [1,2].

Assumption 3. The defect rates of all the products during the rework process is greater than those of the products before the rework process, i.e., $\delta_i \le \gamma_i$ for all $i \in V$. Note that $\delta_i = \gamma_i$ for all $i \in V$ in the current rework network [1,2].

Assumption 4. Even in node α, some defects cannot be sent back to node β for a rework process and must be discarded, i.e., $\delta \le 1$. Note that $\delta = 1$ in the current rework network, which is impractical [1,2].

3.2. Multi-vector structure for the solution

A feasible solution is to identify a successful manner in which b units of flow can be sent into node 1 through which at least d units of flow are received from node n [23–25]. Based on the proposed Assumption 1, a feasible solution must have a rework process only if the total number of perfect products outputted from node n is less than d.

From the proposed Assumption 2, two consecutive processes are required for the rework feasible solution. Hence, two special vectors called non-rework vector and rework vector for the first and second processes are proposed rather than the normal and rework paths used in the current rework network, respectively. The number of rework and non-rework vectors are still growing exponentially with the network size.

Let $p_x = (p_{x,0}, p_{x,1}, p_{x,2}, \ldots, p_{x,n})$ and $q_x = (q_{x,0}, q_{x,1}, q_{x,2}, \ldots, q_{x,n})$ be two non-rework state vectors (without using the rework process), where $p_{x,i}$ and $q_{x,i}$ are the number of perfect and defect products right after processing in node i for $i = 1, 2, \ldots, n$, respectively. For example, in Figs. 2 and 3(a), (14, 14, 12, 11, 10) and (14, 13, 12, 5, 5) are, respectively, two non-rework state vectors of the rework network shown in Fig. 1.

Similarly, let $\pi_y = (\pi_{y,\alpha}, \pi_{y,\beta}, \pi_{y,\beta+1}, \ldots, \pi_{y,n})$ and $\theta_y = (\theta_{y,\alpha}, \theta_{y,\beta}, \theta_{y,\beta+1}, \ldots, \theta_{y,n})$ be two rework state vectors, where $\pi_{y,j}$ and $\theta_{y,j}$ are the number of perfect and defect products right after processing in node i for $i = \beta, \beta+1, \ldots, n$ during the rework process, respectively. Note that $p_{x,0} = b$, $\pi_{y,\alpha} = b^*$, and $q_{x,0} = \theta_{y,0} = 0$. For example, as shown in Fig. 3(b), $\pi_y = (6, 6, 5, 5)$ is a rework state vector of the rework network shown in Fig. 1 and six defects are sent back from node $\alpha = 3$ to node $\beta = 2$ for the rework process, six, five, and five perfect products are outputted from nodes 2, 3, and 4, respectively.

The relationship between p_x and q_x is discussed by lemma 1 and lemma 2 in Appendix A. Similar to Lemma 1, the lemma 3 in Appendix A discusses the relationship between π_y and θ_y.

A multi-vector is a vector formed by vectors [26–28]. To easily

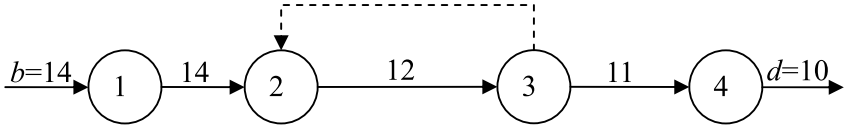


Fig. 2. Feasible solution $X = (p_x, 0)$, where $p_x = (14, 14, 12, 11, 10)$. (a) First process (b) Second process (i.e., rework process).

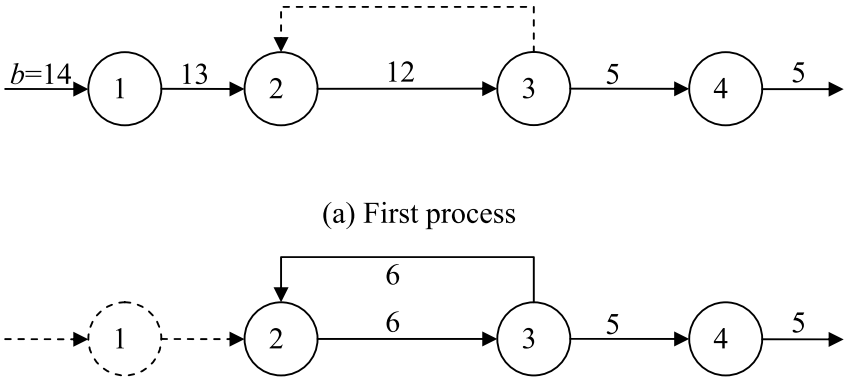


Fig. 3. Feasible solution $X = (p_x, \pi_y)$, where $p_x = (14, 13, 12, 5, 5)$ and $\pi_y = (6, 6, 5, 5)$.

represent the feasible solution, the multi-vector $X = (p_x, \pi_y)$ is used. The feasible solution X is a non-reworked feasible solution if π_y is a zero vector, e.g., $X = (p_x, 0)$, shown in Fig. 2, for $b = 14$, $p_x = (14, 14, 12, 11, 10)$ and $q_x = (0, 0, 2, 1, 1)$. Otherwise, it is a reworked feasible solution, e.g., $X = (p_x, \pi_y)$, shown in Fig. 3, where $b = 14$, $p_x = (14, 13, 12, 5, 5)$, $q_x = (0, 1, 1, 7, 2)$, $\pi_y = (6, 6, 5, 5)$, and $\theta_y = (0, 0, 1, 0)$.

The lemma 4 in Appendix A considers a feasible solution is a vector without the rework process and lemma 5 in Appendix A considers a feasible solution formed by two vectors with the rework process.

4. Proposed 2-STEP 3-DFS algorithm and an example

Based on the innovations [22], the proposed multi-vector concept and assumptions, the 3-DFS algorithm proposed to calculate the reliability of the proposed general rework network is discussed in this section along with examples to demonstrate the related proposed algorithm.

4.1. Proposed 3-DFS concept in find all multi-vectors

To identify all feasible solutions, a 3-DFS is proposed by three depth-first-search methods (DFSs), named DFS_0, DFS_p, and DFS_π. DFS_θ. The DFS_0 is implemented independently to identify all non-reworked feasible solutions; further, it is similar to the traditional DFS, whose details can be found in [19–21], although with some constraints to reduce the size of solution space; DFS_p and DFS_π are implemented in a nested manner to identify the reworked feasible solutions. The proposed 3-DFS is implemented in the following manner:

STEP D0. Implement the DFS_0 to identify all state vectors; assume $p_x = (b, p_{x,1}, p_{x,2}, \ldots, p_{x,n})$ where $d \le p_{x,n}$ and $p_{x,i} \le p_{x,j}$ for all $j \le I$, and let I_n be the set of such state vectors and $I_r = \varnothing$.
STEP D1. Implement the DFS_p to identify a new non-reworked state vector; assume $p_x = (b, p_{x,1}, p_{x,2}, \ldots, p_{x,n})$ for $p_{x,n} < d$ and $p_{x,i} \le p_{x,j}$ for all $j \le i$. If no such vector is identified, then halt.
STEP D2. Implement the DFS_π to identify a new reworked state vector, assume $\pi_y = (\pi_{y,\alpha}, \pi_{y,\beta}, \pi_{y,\beta+1}, \ldots, \pi_{y,n})$, based on p_x, and Lemma 5. If no such vector is identified, then go to STEP D1.
STEP D3. Let $I_r = I_r \cup \{(p_x, \pi_y)\}$ and repeat STEP D2.

The traditional DFS is an implicit enumeration, used to identify all possible feasible solutions. From the above, the proposed nested DFS can also identify all feasible solutions. Hence, we have the lemma 6 shown in Appendix A.

4.2. Occurrent probability of a multi-vector

Any found feasible solution is required to calculate its occurrent probability and the total of all such probability is the required reliability [18,24]. Let the total number of input products be $b = p_{x,0}$, $p_x = (p_{x,0}, p_{x,1}, p_{x,2}, \ldots, p_{x,n})$, $q_x = (q_{x,0}, q_{x,1}, q_{x,2}, \ldots, q_{x,n})$, $\pi_y = (\pi_{y,\alpha}, \pi_{y,\beta}, \pi_{y,\beta+1}, \ldots, \pi_{y,n})$, and $\theta_y = (\theta_{y,\alpha}, \theta_{y,\beta}, \theta_{y,\beta+1}, \ldots, \theta_{y,n})$. Consider the following two cases:

Case 1. The non-rework feasible solution

If (p_x, π_y) is a non-rework feasible solution, i.e., π_y is a zero vector that does not require the rework process, the number of perfect and defect products in node i are $p_{x,i}$ and $q_{x,i} = b - \sum_{k=1}^{i} p_{x,k} = p_{x,i-1} - p_{x,i}$, respectively, from Lemma 2. Therefore, the related occurrence probability of node i is

$$C_{p_{x,i}}^{p_{x,i-1}}(1-\delta_i)^{p_{x,i}}\delta_i^{q_{x,i}}\cdot C_i(p_{x,i-1}), \tag{13}$$

where $C_{p_{x,i}}^{p_{x,i-1}} = \frac{p_{x,i-1}!}{p_{x,i}!\times q_{x,i}!}$ is the total combination to have $p_{x,i}$ and $q_{x,i}$ units of perfect and defect products from node i, δ_i is the defect rate in node i, and $C_i(p_{x,i-1})$ is the probability to have $c_i(p_{x,i-1})$ units of perfect products to input to node i for $i = 1, 2, \ldots, n$. The rework network is constructed in series irrespective of whether it has rework process. Let $p_{x,0} = b$. Thus, the probability of the non-rework feasible solution (p_x, π_y) with $\pi_y = 0$ is

$$Pr((p_x, \pi_y)) = Pr(p_x) = \prod_{i=1}^{n} C_{p_{x,i}}^{p_{x,i-1}}(1-\delta_i)^{p_{x,i}}\delta_i^{q_{x,i}}\cdot C_i(p_{x,i-1}) \tag{14}$$

Case 2. The rework multi-vector

The number of all output perfect products must be larger than or equal to d; otherwise, the number of the output reworked perfect-products via π_y is required to be $(d - p_{w,n})$ at least.

In the rework process, the number of defects sent from node α to node β that required rework is the total number of defects from node α int the first process times a predefined reworked rate, i.e., $b^* = \delta\, q_{y,\alpha}$. Hence, similar to Eq. (14), the occurrent probability of π_y is

$$Pr(\pi_y) = \prod_{i=\beta}^{n} C_{\pi_{y,i}}^{\pi_{y,i-1}}(1-\gamma_i)^{\pi_{y,i}}\gamma_i^{\theta_{y,i}}\cdot C_i(\pi_{y,i-1}), \tag{15}$$

where $\pi_{y,\alpha} = b^*$. The π_y is happen right after p_x to complement the number of perfect products and the probability to have b^* reworked perfect products from $q_{y,\alpha}$ units of defect products is

$Pr(\pi_{y,\alpha}$ units of detective product sent to rework from node α $\mid q_{x,\alpha}$ units of detective product in node α)

$$= Pr(\pi_{y,\alpha} = b^* | q_{x,\alpha}) = C_{\pi_{y,\alpha}}^{q_{x,\alpha}}\delta^{\pi_{y,\alpha}}(1-\delta)^{(q_{x,\alpha}-\pi_{y,\alpha})} \tag{16}$$

Thus,

$$\begin{aligned} &Pr((p_x, \pi_y)) = Pr(p_x) \times C_{\pi_{y,\alpha}}^{q_{x,\alpha}}(1-\delta)^{\pi_{y,\alpha}}\delta^{\theta_{y,\alpha}} \times Pr(\pi_y) \\ &= \prod_{i=1}^{n} C_{p_{x,i}}^{p_{x,i-1}}(1-\delta_i)^{p_{x,i}}\delta_i^{q_{x,i}}\cdot C_i(p_{x,i-1}) \times C_{\pi_{y,\alpha}}^{q_{x,\alpha}}(1-\delta)^{\pi_{y,\alpha}}\delta^{\theta_{y,\alpha}}\cdot \\ &\times \prod_{i=\beta}^{n} C_{\pi_{y,i}}^{\pi_{y,i-1}}(1-\gamma_i)^{\pi_{y,i}}\gamma_i^{\theta_{y,i}}\cdot C_i(\pi_{y,i-1}) \end{aligned} \tag{17}$$

Table 2
Calculation of $Pr((p_x, 0))$.

i	δ_i	$p_{x,i-1}$	$p_{x,i}$	$q_{x,i}$	$C_{p_{x,i}}^{p_{x,i-1}}$	$(1-\delta_i)^{p_{x,i}}$	$\delta_i^{q_{x,i}}$	$c_i(p_{x,i-1})$	$C_i(p_{x,i-1})$	$C_{p_{x,i}}^{p_{x,i=1}}(1-\delta_i)^{p_{x,i}}\delta_i^{q_{x,i}}\cdot C_i(p_{x,i-1})$
1	0.05	14	14	0	1	0.487675	1	15	0.93	0.453537731
2	0.10	14	12	2	91	0.282430	0.01	15	0.95	0.244160334
3	0.15	12	11	1	12	0.167343	0.15	12	0.01	0.003012178
4	0.20	11	10	1	11	0.107374	0.2	12	0.015	0.003543348

Table 3
Calculation of Pr(p_x).

i	δ_i	$p_{x,i-1}$	$p_{x,i}$	$q_{x,i}$	$C_{p_{x,i}}^{p_{x,i-1}}$	$(1-\delta_i)^{p_{x,i}}$	$\delta_i^{q_{x,i}}$	$c_i(p_{x,i-1})$	$C_i(p_{x,i-1})$	$C_{p_{x,i}}^{p_{x,i-1}}(1-\delta_i)^{p_{x,i}}\delta_i^{q_{x,i}}\cdot C_i(p_{x,i-1})$
1	0.05	14	13	1	14	0.513342	0.05	15	0.93	0.334185696
2	0.10	13	12	1	13	0.282430	0.10	15	0.95	0.348800478
3	0.15	12	5	7	792	0.443705	1.70859E-06	12	0.01	6.00425E-06
4	0.20	5	5	0	1	0.327680	1.00	12	0.015	0.004915200
									Product	3.44005E-09

Table 4
Calculation of Pr(π_y).

i	γ_i	$\pi_{y,i-1}$	$\pi_{y,i}$	$\theta_{y,i}$	$C_{\pi_{y,i}}^{\pi_{y,i-1}}$	$(1-\gamma_i)^{\pi_{y,i}}$	$\gamma_i^{\theta_{y,i}}$	$c_i(\pi_{y,i-1})$	$C_i(\pi_{y,i-1})$	$C_{\pi_{y,i}}^{\pi_{y,i-1}}(1-\gamma_i)^{\pi_{y,i}}\gamma_i^{\theta_{y,i}}\cdot C_i(\pi_{y,i-1})$
1*	0.06	6	6	0	1	0.689870	1.00	10	0.050	0.034493489
2	0.12	6	5	1	6	0.527732	0.12	10	0.020	0.007599340
3	0.18	5	5	0	1	0.370740	1.00	6	0.010	0.003707398
4	0.24	5	5	0	1	0.253553	1.00	6	0.005	0.001267763
									Product	1.23203E-09

* Nodes 3 and 2 are nodes α and β in Fig. 3, respectively.

Table 5
Occurrent probability of each state of Example 1.

Node State	1	2	3	4
0	0.010000	0.010000	0.005000	0.005000
1	0.010000	0.020000	0.005000	0.005000
2	0.010000	0.020000	0.005000	0.005000
3	0.010000	0.020000	0.005000	0.005000
4	0.010000	0.020000	0.010000	0.005000
5	0.010000	0.020000	0.010000	0.005000
6	0.050000	0.020000	0.010000	0.005000
7	0.050000	0.020000	0.010000	0.010000
8	0.050000	0.020000	0.010000	0.010000
9	0.050000	0.020000	0.010000	0.010000
10	0.050000	0.020000	0.010000	0.015000
11	0.930000	0.950000	0.010000	0.015000
12	0.930000	0.950000	0.010000	0.015000
13	0.930000	0.950000	0.960000	0.960000
14	0.930000	0.950000	0.960000	0.960000

$$= \frac{b!}{q_{x,1}!\cdot q_{x,2}!\cdots q_{x,n}!}\prod_{i=1}^{n}(1-\delta_i)^{p_{x,i}}\delta_i^{q_{x,i}}\cdot C_i(p_{x,i-1}) \times C_{\pi_{y,\alpha}}^{q_{x,\alpha}}(1-\delta)^{\pi_{y,\alpha}}\delta^{\theta_{y,\alpha}} \times \frac{q_{x,\alpha}!}{\theta_{y,\alpha}!\cdot\theta_{y,\beta}!\cdot\theta_{y,\beta+1}!\cdots\theta_{y,n}!}\prod_{i=\beta}^{n}(1-\gamma_i)^{\pi_{y,i}}\gamma_i^{\theta_{y,i}}\cdot C_i(\pi_{y,i-1}) \quad (18)$$

From the above, we have the lemma 7 given in Appendix A.

For example, let $\delta_1=0.05$, $\delta_2=0.10$, $\delta_3=0.15$, $\delta_4=0.20$, $\delta=0.20$, $\gamma_1=0.06$, $\gamma_2=0.10$, $\gamma_3=0.15$, and $\gamma_4=0.20$. In Fig. 2, X = Pr((p_x, 0)) is a feasible solution, where p_x = (14, 14, 12, 11, 10) and we have Table 2:

Hence, Pr(X) = 0.453537731 × 0.244160334 × 0.003012178 × 0.003543348 = 1.18191E-06.

In Fig. 3, X = (p_x, π_y) is a feasible solution, where p_x = (14, 13, 12, 5, 5) and π_y = (6, 6, 5, 5). Similarly, we have the following results of 1st process and 2nd process in calculating Pr(X) as summarized in Tables 3 and 4, respectively:

Table 5.

Hence,

$$Pr(\pi_{y,\alpha}=6|q_{x,\alpha}=7) = C_6^7(1-\delta)^6\delta^1 = 0.531441$$
$$Pr(X) = Pr((p_x, \pi_y))$$
$$= Pr(p_x) \times Pr(\pi_{y,\alpha}=6|q_{x,\alpha}=7) \times Pr(\pi_y)$$
$$= 0.334185696 \times 0.348800478 \times 6.00425E-06 \times 0.004915200 \times 0.531441 \times 0.034493489 \times 0.007599340 \times 0.003707398 \times 0.001267763$$
$$= 2.25E-18$$

4.3. Proposed algorithm

The algorithm proposed to estimate the general rework reliability problem is discussed completely in this subsection based on Sections 4.1 and 4.2.

STEP 1. Implement STEPs D0–D3 in the proposed 3-DFS discussed in Section 4.1 to estimate all rework multi-vectors.
STEP 2. Calculate the occurrent probability of all feasible solutions (events) obtained in STEP 1.
STEP 3. Sum up all the occurrent probabilities calculated in SETP 2 and the result is the final reliability required.

From Lemmas 1–7, we have the following theorem:
Theorem 1. *The proposed algorithm finds all feasible solutions and can calculate the reliability of the general rework network.*

5. Computational experiments

Before calculating the exact reliability of a general rework network, all feasible solutions must first be obtained [1,2,20,21]. However, all feasible solutions possess a computational difficulty [17], in the worst case, it raises exponentially with size of the network due to the characteristic of the NP-hard problem [16,25]. Owing to this inherent problem, a moderate size network shown in Fig. 1 was selected to

Table 6
Final results obtained from the low defect rate setting of Fig. 1.

B	d	N_n	N_r	R_n	R_r	$R_l = R_n + R_r$	R_n/R_l	R_r/R_l
1	1	1	1	4.6588E-09	5.4590E-17	4.6588E-09	0.99999999	1.1718E-08
2	1	5	4	4.9767E-09	7.4500E-18	4.9767E-09	1.00000000	1.4970E-09
	2	1	2	4.3408E-09	1.0291E-16	4.3408E-09	0.99999998	2.3707E-08
3	1	15	10	4.9984E-09	7.6000E-19	4.9984E-09	1.00000000	1.5205E-10
	2	5	8	4.9333E-09	2.1040E-17	4.9333E-09	1.00000000	4.2649E-09
	3	1	3	4.0446E-09	1.4553E-16	4.0446E-09	0.99999996	3.5981E-08
4	1	35	20	9.5211E-09	9.0000E-20	9.5211E-09	1.00000000	9.4527E-12
	2	15	20	9.5137E-09	4.0300E-18	9.5137E-09	1.00000000	4.2360E-10
	3	5	12	9.3433E-09	6.4790E-17	9.3433E-09	0.99999999	6.9344E-09
	4	1	4	7.5372E-09	3.6585E-16	7.5372E-09	0.99999995	4.8539E-08
5	1	70	35	9.9706E-09	1.0000E-20	9.9706E-09	1.00000000	1.0029E-12
	2	35	40	9.9698E-09	5.8000E-19	9.9698E-09	1.00000000	5.8175E-11
	3	15	30	9.9464E-09	1.2620E-17	9.9464E-09	1.00000000	1.2688E-09
	4	5	16	9.5946E-09	1.2439E-16	9.5946E-09	0.99999999	1.2965E-08
	5	1	5	7.0228E-09	4.3116E-16	7.0228E-09	0.99999994	6.1394E-08
6	1	126	56	4.9993E-08	0.0000E+00	4.9993E-08	1.00000000	0.0000E+00
	2	70	70	4.9992E-08	3.4000E-19	4.9992E-08	1.00000000	6.8010E-12
	3	35	60	4.9979E-08	9.3500E-18	4.9979E-08	1.00000000	1.8708E-10
	4	15	40	4.9728E-08	1.2719E-16	4.9728E-08	1.00000000	2.5577E-09
	5	5	20	4.7096E-08	8.7853E-16	4.7096E-08	0.99999998	1.8654E-08
	6	1	6	3.2718E-08	2.4391E-15	3.2718E-08	0.99999993	7.4551E-08
7	1	210	84	8.6396E-08	0.0000E+00	8.6396E-08	1.00000000	0.0000E+00
	2	126	112	8.6396E-08	4.0000E-20	8.6396E-08	1.00000000	4.6299E-13
	3	70	105	8.6394E-08	1.2500E-18	8.6394E-08	1.00000000	1.4469E-11
	4	35	80	8.6363E-08	2.2630E-17	8.6363E-08	1.00000000	2.6203E-10
	5	15	50	8.5926E-08	2.3538E-16	8.5926E-08	1.00000000	2.7393E-09
	6	5	24	8.2071E-08	1.5594E-15	8.2071E-08	0.99999998	1.9001E-08
	7	1	7	6.0970E-08	5.3664E-15	6.0970E-08	0.99999991	8.8018E-08
8	1	330	120	9.7696E-08	0.0000E+00	9.7696E-08	1.00000000	0.0000E+00
	2	210	168	9.7696E-08	0.0000E+00	9.7696E-08	1.00000000	0.0000E+00
	3	126	168	9.7696E-08	1.6000E-19	9.7696E-08	1.00000000	1.6377E-12
	4	70	140	9.7692E-08	3.7300E-18	9.7692E-08	1.00000000	3.8181E-11
	5	35	100	9.7628E-08	5.1730E-17	9.7628E-08	1.00000000	5.2987E-10
	6	15	60	9.6827E-08	4.9693E-16	9.6827E-08	0.99999999	5.1322E-09
	7	5	28	9.0096E-08	2.9094E-15	9.0096E-08	0.99999997	3.2292E-08
	8	1	8	5.6809E-08	5.7833E-15	5.6809E-08	0.99999990	1.0180E-07
9	1	495	165	9.9701E-08	0.0000E+00	9.9701E-08	1.00000000	0.0000E+00
	2	330	240	9.9701E-08	0.0000E+00	9.9701E-08	1.00000000	0.0000E+00
	3	210	252	9.9701E-08	1.0000E-20	9.9701E-08	1.00000000	1.0030E-13
	4	126	224	9.9700E-08	5.6000E-19	9.9700E-08	1.00000000	5.6168E-12
	5	70	175	9.9691E-08	9.7100E-18	9.9691E-08	1.00000000	9.7400E-11
	6	35	120	9.9554E-08	1.1604E-16	9.9554E-08	1.00000000	1.1656E-09
	7	15	70	9.8046E-08	8.9156E-16	9.8046E-08	0.99999999	9.0932E-09
	8	5	32	8.7824E-08	3.5235E-15	8.7824E-08	0.99999996	4.0120E-08
	9	1	9	5.2932E-08	6.1355E-15	5.2932E-08	0.99999988	1.1591E-07
10	1	715	220	1.3173E-07	0.0000E+00	1.3173E-07	1.00000000	0.0000E+00
	2	495	330	1.3173E-07	0.0000E+00	1.3173E-07	1.00000000	0.0000E+00
	3	330	360	1.3173E-07	0.0000E+00	1.3173E-07	1.00000000	0.0000E+00
	4	210	336	1.3173E-07	7.0000E-20	1.3173E-07	1.00000000	5.3138E-13
	5	126	280	1.3173E-07	1.5900E-18	1.3173E-07	1.00000000	1.2070E-11
	6	70	210	1.3171E-07	2.2780E-17	1.3171E-07	1.00000000	1.7296E-10
	7	35	140	1.3144E-07	2.1804E-16	1.3144E-07	1.00000000	1.6589E-09
	8	15	80	1.2906E-07	1.2735E-15	1.2906E-07	0.99999999	9.8675E-09
	9	5	36	1.1643E-07	4.8630E-15	1.1643E-07	0.99999996	4.1769E-08
	10	1	10	7.3980E-08	9.6439E-15	7.3980E-08	0.99999987	1.3036E-07
11	1	1001	286	1.1679E-04	0.0000E+00	1.1679E-04	1.00000000	0.0000E+00
	2	715	440	1.1679E-04	0.0000E+00	1.1679E-04	1.00000000	0.0000E+00
	3	495	495	1.1679E-04	0.0000E+00	1.1679E-04	1.00000000	0.0000E+00
	4	330	480	1.1679E-04	2.0100E-18	1.1679E-04	1.00000000	1.7211E-14
	5	210	420	1.1679E-04	9.5000E-17	1.1679E-04	1.00000000	8.1345E-13
	6	126	336	1.1679E-04	1.8210E-15	1.1679E-04	1.00000000	1.5593E-11
	7	70	245	1.1677E-04	2.4261E-14	1.1677E-04	1.00000000	2.0777E-10
	8	35	160	1.1654E-04	2.1163E-13	1.1654E-04	1.00000000	1.8159E-09
	9	15	90	1.1458E-04	1.3312E-12	1.1458E-04	0.99999999	1.1618E-08
	10	5	40	1.0293E-04	5.4140E-12	1.0293E-04	0.99999995	5.2600E-08
	11	1	11	6.0901E-05	8.8396E-12	6.0901E-05	0.99999985	1.4515E-07

(*continued on next page*)

Table 6 (*continued*)

B	d	N_n	N_r	R_n	R_r	$R_l = R_n + R_r$	R_n/R_l	R_r/R_l
12	1	1365	364	1.3118E-04	0.0000E+00	1.3118E-04	1.00000000	0.0000E+00
	2	1001	572	1.3118E-04	0.0000E+00	1.3118E-04	1.00000000	0.0000E+00
	3	715	660	1.3118E-04	0.0000E+00	1.3118E-04	1.00000000	0.0000E+00
	4	495	660	1.3118E-04	0.0000E+00	1.3118E-04	1.00000000	0.0000E+00
	5	330	600	1.3118E-04	2.0000E-17	1.3118E-04	1.00000000	1.5246E-13
	6	210	504	1.3118E-04	4.6000E-16	1.3118E-04	1.00000000	3.5066E-12
	7	126	392	1.3118E-04	6.8800E-15	1.3118E-04	1.00000000	5.2448E-11
	8	70	280	1.3112E-04	6.8856E-14	1.3112E-04	1.00000000	5.2515E-10
	9	35	180	1.3052E-04	5.0382E-13	1.3052E-04	1.00000000	3.8601E-09
	10	15	100	1.2629E-04	2.5291E-12	1.2629E-04	0.99999998	2.0026E-08
	11	5	44	1.0662E-04	7.1586E-12	1.0662E-04	0.99999993	6.7142E-08
	12	1	12	5.6744E-05	9.0956E-12	5.6744E-05	0.99999984	1.6029E-07
13	1	1820	455	4.5361E-01	0.0000E+00	4.5361E-01	1.00000000	0.0000E+00
	2	1365	728	4.5361E-01	0.0000E+00	4.5361E-01	1.00000000	0.0000E+00
	3	1001	858	4.5361E-01	0.0000E+00	4.5361E-01	1.00000000	0.0000E+00
	4	715	880	4.5361E-01	0.0000E+00	4.5361E-01	1.00000000	0.0000E+00
	5	495	825	4.5361E-01	0.0000E+00	4.5361E-01	1.00000000	0.0000E+00
	6	330	720	4.5361E-01	0.0000E+00	4.5361E-01	1.00000000	0.0000E+00
	7	210	588	4.5361E-01	5.5012E-14	4.5361E-01	1.00000000	1.2127E-13
	8	126	448	4.5361E-01	1.1130E-12	4.5361E-01	1.00000000	2.4536E-12
	9	70	315	4.5361E-01	2.4161E-11	4.5361E-01	1.00000000	5.3264E-11
	10	35	200	4.5349E-01	3.8042E-10	4.5349E-01	1.00000000	8.3886E-10
	11	15	110	4.5180E-01	2.4999E-09	4.5180E-01	0.99999999	5.5333E-09
	12	5	48	4.3453E-01	1.6981E-08	4.3453E-01	0.99999996	3.9080E-08
	13	1	13	3.2485E-01	5.7107E-08	3.2485E-01	0.99999982	1.7580E-07
14	1	2380	560	7.1278E-01	0.0000E+00	7.1278E-01	1.00000000	0.0000E+00
	2	1820	910	7.1278E-01	0.0000E+00	7.1278E-01	1.00000000	0.0000E+00
	3	1365	1092	7.1278E-01	0.0000E+00	7.1278E-01	1.00000000	0.0000E+00
	4	1001	1144	7.1278E-01	0.0000E+00	7.1278E-01	1.00000000	0.0000E+00
	5	715	1100	7.1278E-01	0.0000E+00	7.1278E-01	1.00000000	0.0000E+00
	6	495	990	7.1278E-01	0.0000E+00	7.1278E-01	1.00000000	0.0000E+00
	7	330	840	7.1278E-01	2.9976E-14	7.1278E-01	1.00000000	4.2055E-14
	8	210	672	7.1278E-01	7.2597E-13	7.1278E-01	1.00000000	1.0185E-12
	9	126	504	7.1278E-01	1.6065E-11	7.1278E-01	1.00000000	2.2539E-11
	10	70	350	7.1270E-01	2.6518E-10	7.1270E-01	1.00000000	3.7207E-10
	11	35	220	7.1154E-01	2.0163E-09	7.1154E-01	1.00000000	2.8336E-09
	12	15	120	6.9906E-01	1.3408E-08	6.9906E-01	0.99999998	1.9180E-08
	13	5	52	6.1304E-01	5.3848E-08	6.1304E-01	0.99999991	8.7838E-08
	14	1	14	3.0268E-01	5.8016E-08	3.0268E-01	0.99999981	1.9168E-07
15	1	3060	680	7.9285E-01	0.0000E+00	7.9285E-01	1.00000000	0.0000E+00
	2	2380	1120	7.9285E-01	0.0000E+00	7.9285E-01	1.00000000	0.0000E+00
	3	1820	1365	7.9285E-01	0.0000E+00	7.9285E-01	1.00000000	0.0000E+00
	4	1365	1456	7.9285E-01	0.0000E+00	7.9285E-01	1.00000000	0.0000E+00
	5	1001	1430	7.9285E-01	0.0000E+00	7.9285E-01	1.00000000	0.0000E+00
	6	715	1320	7.9285E-01	0.0000E+00	7.9285E-01	1.00000000	0.0000E+00
	7	495	1155	7.9285E-01	7.1054E-15	7.9285E-01	1.00000000	8.9619E-15
	8	330	960	7.9285E-01	2.5402E-13	7.9285E-01	1.00000000	3.2039E-13
	9	210	756	7.9285E-01	5.8811E-12	7.9285E-01	1.00000000	7.4177E-12
	10	126	560	7.9282E-01	1.0050E-10	7.9282E-01	1.00000000	1.2676E-10
	11	70	385	7.9239E-01	8.7117E-10	7.9239E-01	1.00000000	1.0994E-09
	12	35	240	7.8747E-01	6.0260E-09	7.8747E-01	0.99999999	7.6524E-09
	13	15	130	7.5071E-01	2.7440E-08	7.5071E-01	0.99999996	3.6552E-08
	14	5	56	5.9186E-01	5.8556E-08	5.9186E-01	0.99999990	9.8936E-08
	15	1	15	2.8202E-01	5.8643E-08	2.8202E-01	0.99999979	2.0794E-07

demonstrate this methodology instead of presenting practically large network systems.

Consider the general rework network in Fig. 1 of Section 1, where nodes 1, 2, 3, and 4 are the source node, the input node of the rework process (i.e., $\alpha = 3$), first node of the rework process (i.e., $\beta = 3$), and sink nodes, respectively. Nodes 0 and 5 are two artificial nodes to represent the input and output number of products in the general rework network. The proposed algorithm is used to obtain all feasible solutions to input $b = 1, 2, \ldots, 15$ units of products and output at least $d = 1, 2, \ldots, b$ perfect products after processing in the general rework network from nodes 1 to 4. The maximal capacity of each node is 15 and the reliability is zero if b is larger than 15.

To calculate Eq. (16) and Eq. (18) efficiently, the first step in the procedure is to list all occurrent probability of each state as follows:

The second step is to implement the proposed 3-DFS to obtain all feasible solutions, then the summations of these occurrent probabilities based on Eq. (16) or Eq. (18) provides final reliability. The final results of calculating the reliability using the proposed 3-DFS are listed in Tables 6 and 7 in Appendix B for lower and higher defect rate settings: (δ_1, δ_2, δ_3, δ_4, δ, γ_1, γ_2, γ_3, γ_4) = (0.010, 0.015, 0.020, 0.025, 0.02, 0.012, 0.018, 0.022, 0.029), and (0.05, 0.10, 0.15, 0.20, 0.20, 0.06, 0.12, 0.18, 0.24), respectively.

Table 7
The final results obtained from the higher defect rate setting of Fig. 1.

b	d	N_n	N_r	R_n	R_r	$R_h = R_n + R_r$	R_n/R_h	R_r/R_h	R_h/R_l
1	1	1	1	2.9070E-09	1.3347E-16	2.9070E-09	0.99999995	4.5913E-08	1.602607
2	1	5	4	4.1239E-09	1.9555E-16	4.1239E-09	0.99999997	4.7419E-08	1.206807
	2	1	2	1.6901E-09	8.3800E-17	1.6901E-09	0.99999988	4.9582E-08	2.568350
3	1	15	10	4.6333E-09	2.4492E-16	4.6333E-09	0.99999998	5.2861E-08	1.078813
	2	5	8	3.1051E-09	1.8005E-16	3.1051E-09	0.99999992	5.7985E-08	1.588776
	3	1	3	9.8264E-10	6.1200E-17	9.8264E-10	0.99999982	6.2281E-08	4.116056
4	1	35	20	7.4904E-09	2.9253E-16	7.4904E-09	0.99999999	3.9054E-08	1.271098
	2	15	20	6.3994E-09	4.9571E-16	6.3994E-09	0.99999995	7.7462E-08	1.486650
	3	5	12	3.8634E-09	2.9507E-16	3.8634E-09	0.99999987	7.6377E-08	2.418445
	4	1	4	1.1426E-09	4.0750E-17	1.1426E-09	0.99999974	3.5664E-08	6.596421
5	1	70	35	9.1295E-09	2.5059E-16	9.1295E-09	1.00000000	2.7448E-08	1.092133
	2	35	40	8.4291E-09	5.8979E-16	8.4291E-09	0.99999997	6.9971E-08	1.182788
	3	15	30	6.2929E-09	6.1375E-16	6.2929E-09	0.99999991	9.7531E-08	1.580577
	4	5	16	3.0558E-09	2.2699E-16	3.0558E-09	0.99999980	7.4281E-08	3.139804
	5	1	5	6.6432E-10	1.2084E-16	6.6432E-10	0.99999966	1.8190E-07	10.571472
6	1	126	56	4.8695E-08	8.8507E-16	4.8695E-08	1.00000000	1.8176E-08	1.026648
	2	70	70	4.6711E-08	2.6108E-15	4.6711E-08	0.99999998	5.5892E-08	1.070256
	3	35	60	3.9406E-08	3.8510E-15	3.9406E-08	0.99999993	9.7727E-08	1.268331
	4	15	40	2.5290E-08	2.8296E-15	2.5290E-08	0.99999985	1.1189E-07	1.966337
	5	5	20	1.0274E-08	8.3956E-16	1.0274E-08	0.99999972	8.1720E-08	4.584119
	6	1	6	1.9312E-09	6.6500E-17	1.9312E-09	0.99999957	3.4435E-08	16.941917
7	1	210	84	5.4984E-08	5.6785E-16	5.4984E-08	1.00000000	1.0327E-08	1.571278
	2	126	112	5.3950E-08	2.0456E-15	5.3950E-08	0.99999999	3.7916E-08	1.601409
	3	70	105	4.9484E-08	4.0243E-15	4.9484E-08	0.99999996	8.1326E-08	1.745917
	4	35	80	3.8846E-08	4.6527E-15	3.8846E-08	0.99999990	1.1977E-07	2.223213
	5	15	50	2.3565E-08	3.2738E-15	2.3565E-08	0.99999980	1.3893E-07	3.646289
	6	5	24	9.8691E-09	1.2097E-15	9.8691E-09	0.99999967	1.2257E-07	8.315923
	7	1	7	2.2456E-09	3.5150E-17	2.2456E-09	0.99999946	1.5653E-08	27.151237
8	1	330	120	6.5488E-08	3.4823E-16	6.5488E-08	1.00000000	5.3174E-09	1.491803
	2	210	168	6.4974E-08	1.5170E-15	6.4974E-08	0.99999999	2.3348E-08	1.503612
	3	126	168	6.2400E-08	3.8212E-15	6.2400E-08	0.99999998	6.1237E-08	1.565628
	4	70	140	5.4974E-08	6.0594E-15	5.4974E-08	0.99999993	1.1022E-07	1.777059
	5	35	100	4.1174E-08	6.4095E-15	4.1174E-08	0.99999985	1.5567E-07	2.371085
	6	15	60	2.3738E-08	4.0151E-15	2.3738E-08	0.99999973	1.6914E-07	4.078963
	7	5	28	8.8255E-09	8.5538E-16	8.8255E-09	0.99999955	9.6921E-08	10.208548
	8	1	8	1.3056E-09	1.8230E-17	1.3056E-09	0.99999934	1.3963E-08	43.512766
9	1	495	165	7.6755E-08	2.0880E-16	7.6755E-08	1.00000000	2.7204E-09	1.298950
	2	330	240	7.6505E-08	1.0846E-15	7.6505E-08	1.00000000	1.4177E-08	1.303189
	3	210	252	7.5067E-08	3.3490E-15	7.5067E-08	0.99999999	4.4614E-08	1.328156
	4	126	224	7.0122E-08	6.6440E-15	7.0122E-08	0.99999995	9.4750E-08	1.421817
	5	70	175	5.8733E-08	8.9587E-15	5.8734E-08	0.99999989	1.5253E-07	1.697353
	6	35	120	4.0331E-08	7.5576E-15	4.0331E-08	0.99999978	1.8739E-07	2.468436
	7	15	70	1.9843E-08	3.1922E-15	1.9843E-08	0.99999962	1.6087E-07	4.941111
	8	5	32	5.6777E-09	5.9657E-16	5.6777E-09	0.99999944	1.0507E-07	15.468339
	9	1	9	7.5906E-10	9.3700E-18	7.5906E-10	0.99999921	1.2344E-08	69.733869
10	1	715	220	8.8007E-08	1.2324E-16	8.8007E-08	1.00000000	1.4003E-09	1.496841
	2	495	330	8.7887E-08	7.4775E-16	8.7887E-08	1.00000000	8.5081E-09	1.498886
	3	330	360	8.7097E-08	2.7217E-15	8.7097E-08	0.99999999	3.1249E-08	1.512477
	4	210	336	8.3922E-08	6.4031E-15	8.3922E-08	0.99999997	7.6299E-08	1.569700
	5	126	280	7.5214E-08	1.0333E-14	7.5214E-08	0.99999991	1.3738E-07	1.751423
	6	70	210	5.8309E-08	1.0972E-14	5.8309E-08	0.99999982	1.8818E-07	2.258821
	7	35	140	3.5485E-08	7.0890E-15	3.5485E-08	0.99999969	1.9977E-07	3.703966
	8	15	80	1.5306E-08	2.8938E-15	1.5306E-08	0.99999954	1.8906E-07	8.431914
	9	5	36	4.3910E-09	6.1768E-16	4.3910E-09	0.99999934	1.4067E-07	26.514404
	10	1	10	6.6197E-10	1.4266E-15	6.6198E-10	0.99999907	2.1551E-06	111.755993
11	1	1001	286	5.4447E-05	2.4166E-14	5.4447E-05	1.00000000	4.4384E-10	2.144970
	2	715	440	5.4431E-05	1.8803E-13	5.4431E-05	1.00000000	3.4546E-09	2.145596
	3	495	495	5.4296E-05	8.6854E-13	5.4296E-05	1.00000000	1.5996E-08	2.150932
	4	330	480	5.3590E-05	2.5921E-12	5.3590E-05	0.99999998	4.8369E-08	2.179266
	5	210	420	5.1084E-05	5.3501E-12	5.1084E-05	0.99999995	1.0473E-07	2.286153
	6	126	336	4.4844E-05	7.6519E-12	4.4844E-05	0.99999988	1.7063E-07	2.604232
	7	70	245	3.3924E-05	7.5005E-12	3.3924E-05	0.99999977	2.2109E-07	3.441984
	8	35	160	2.0622E-05	5.1480E-12	2.0622E-05	0.99999964	2.4964E-07	5.651206
	9	15	90	9.3522E-06	2.1930E-12	9.3522E-06	0.99999945	2.3449E-07	12.251617
	10	5	40	2.7182E-06	3.7369E-13	2.7182E-06	0.99999919	1.3748E-07	37.866728
	11	1	11	3.4003E-07	1.4482E-15	3.4003E-07	0.99999890	4.2590E-09	179.100943

(*continued on next page*)

Table 7 (*continued*)

b	d	N_n	N_r	R_n	R_r	$R_h = R_n + R_r$	R_n/R_h	R_r/R_h	R_h/R_l
12	1	1365	364	8.9657E-05	2.6125E-14	8.9657E-05	1.00000000	2.9139E-10	1.463165
	2	1001	572	8.9641E-05	2.1477E-13	8.9641E-05	1.00000000	2.3959E-09	1.463423
	3	715	660	8.9499E-05	1.0530E-12	8.9499E-05	1.00000000	1.1766E-08	1.465735
	4	495	660	8.8717E-05	3.3729E-12	8.8717E-05	0.99999999	3.8019E-08	1.478656
	5	330	600	8.5772E-05	7.5774E-12	8.5772E-05	0.99999996	8.8344E-08	1.529431
	6	210	504	7.7936E-05	1.2079E-11	7.7936E-05	0.99999990	1.5499E-07	1.683202
	7	126	392	6.3054E-05	1.3661E-11	6.3054E-05	0.99999981	2.1666E-07	2.080412
	8	70	280	4.2791E-05	1.1205E-11	4.2791E-05	0.99999968	2.6184E-07	3.064113
	9	35	180	2.2807E-05	6.1485E-12	2.2807E-05	0.99999951	2.6959E-07	5.722868
	10	15	100	8.5751E-06	1.8568E-12	8.5751E-06	0.99999928	2.1653E-07	14.727093
	11	5	44	1.9058E-06	2.5432E-13	1.9058E-06	0.99999903	1.3345E-07	55.945032
	12	1	12	1.9770E-07	8.6499E-14	1.9770E-07	0.99999871	4.3753E-07	287.028429
13	1	1820	455	1.4329E-02	2.2470E-13	1.4329E-02	1.00000000	1.5681E-11	31.656033
	2	1365	728	1.4329E-02	1.4669E-12	1.4329E-02	1.00000000	1.0237E-10	31.656079
	3	1001	858	1.4329E-02	5.7525E-12	1.4329E-02	1.00000000	4.0145E-10	31.656625
	4	715	880	1.4327E-02	2.8621E-11	1.4327E-02	1.00000000	1.9977E-09	31.660863
	5	495	825	1.4316E-02	1.2093E-10	1.4316E-02	1.00000000	8.4470E-09	31.684772
	6	330	720	1.4269E-02	4.1427E-10	1.4269E-02	0.99999999	2.9033E-08	31.789874
	7	210	588	1.4101E-02	1.1507E-09	1.4101E-02	0.99999997	8.1605E-08	32.169235
	8	126	448	1.3605E-02	2.4971E-09	1.3605E-02	0.99999992	1.8355E-07	33.342711
	9	70	315	1.2399E-02	3.8569E-09	1.2399E-02	0.99999980	3.1108E-07	36.585199
	10	35	200	1.0060E-02	3.9747E-09	1.0060E-02	0.99999962	3.9511E-07	45.079821
	11	15	110	6.6155E-03	2.8966E-09	6.6155E-03	0.99999940	4.3785E-07	68.293593
	12	5	48	3.0333E-03	1.0582E-09	3.0333E-03	0.99999905	3.4885E-07	143.250067
	13	1	13	7.0619E-04	3.9920E-13	7.0620E-04	0.99999850	5.6528E-10	459.993765
14	1	2380	560	6.2264E-02	1.5070E-12	6.2264E-02	1.00000000	2.4203E-11	11.447697
	2	1820	910	6.2264E-02	5.5998E-12	6.2264E-02	1.00000000	8.9936E-11	11.447703
	3	1365	1092	6.2264E-02	1.8300E-11	6.2264E-02	1.00000000	2.9391E-10	11.447783
	4	1001	1144	6.2260E-02	8.6079E-11	6.2260E-02	1.00000000	1.3826E-09	11.448502
	5	715	1100	6.2235E-02	3.9494E-10	6.2235E-02	1.00000000	6.3460E-09	11.453166
	6	495	990	6.2107E-02	1.4774E-09	6.2107E-02	0.99999999	2.3788E-08	11.476660
	7	330	840	6.1592E-02	4.4128E-09	6.1592E-02	0.99999998	7.1646E-08	11.572662
	8	210	672	5.9896E-02	1.0134E-08	5.9896E-02	0.99999993	1.6920E-07	11.900320
	9	126	504	5.5406E-02	1.6517E-08	5.5406E-02	0.99999982	2.9810E-07	12.864665
	10	70	350	4.6108E-02	1.8129E-08	4.6108E-02	0.99999964	3.9318E-07	15.457368
	11	35	220	3.1653E-02	1.4067E-08	3.1653E-02	0.99999943	4.4440E-07	22.479387
	12	15	120	1.5757E-02	6.1265E-09	1.5757E-02	0.99999911	3.8880E-07	44.363767
	13	5	52	4.5492E-03	7.1386E-10	4.5492E-03	0.99999865	1.5692E-07	134.758684
	14	1	14	4.1058E-04	9.8702E-13	4.1058E-04	0.99999826	2.4039E-09	737.189202
15	1	3060	680	1.5129E-01	5.1850E-12	1.5129E-01	1.00000000	3.4273E-11	5.240724
	2	2380	1120	1.5129E-01	1.6350E-11	1.5129E-01	1.00000000	1.0807E-10	5.240726
	3	1820	1365	1.5129E-01	4.2000E-11	1.5129E-01	1.00000000	2.7762E-10	5.240745
	4	1365	1456	1.5128E-01	1.6718E-10	1.5128E-01	1.00000000	1.1051E-09	5.240936
	5	1001	1430	1.5124E-01	7.6874E-10	1.5124E-01	1.00000000	5.0829E-09	5.242292
	6	715	1320	1.5103E-01	3.0206E-09	1.5103E-01	0.99999999	2.0001E-08	5.249771
	7	495	1155	1.5008E-01	9.4572E-09	1.5008E-01	0.99999998	6.3016E-08	5.282971
	8	330	960	1.4672E-01	2.2639E-08	1.4672E-01	0.99999994	1.5430E-07	5.403878
	9	210	756	1.3730E-01	3.8655E-08	1.3730E-01	0.99999984	2.8153E-07	5.774505
	10	126	560	1.1686E-01	4.5134E-08	1.1686E-01	0.99999967	3.8621E-07	6.784198
	11	70	385	8.3667E-02	3.7545E-08	8.3667E-02	0.99999946	4.4874E-07	9.470721
	12	35	240	4.5222E-02	1.9164E-08	4.5222E-02	0.99999917	4.2378E-07	17.413409
	13	15	130	1.5810E-02	4.4246E-09	1.5810E-02	0.99999879	2.7987E-07	47.483897
	14	5	56	2.8168E-03	1.5943E-16	2.8168E-03	0.99999843	5.6601E-14	210.120915
	15	1	15	2.3871E-04	1.3347E-16	2.3871E-04	0.99999799	5.5912E-13	1181.42449

From Tables 6 and 7, we draw the following observations:

1 Irrespective of the values of δ_i, δ, and γ_i for $i = 1, 2, 3, 4$, the numbers of N_n and N_r both are always constants, e.g., $N_n = 35$ for $(b, d) = (4, 1)$ and $N_r = 35$ for $(b, d) = (5, 2)$ in Tables 6 and 7.
2 Because a lower defect rate has a higher chance for lesser number of defects than that of the higher defect rate, the setting with the lower defect rate has better R_n, R_r, and R than those of the setting with a higher defect rate. The above observation is confirmed from the ratio of R_h/R_l, which is always larger than 1, listed in the last column in Table 7.

The value of R_h/R_l is increasing with the increment of values b and d, respectively. Surprisingly, the value of R_h/R_l is up to 1181.42449 for $b = d = 15$. For example, Fig. 4 is the 3D surface plot drawn by b, d and the final calculation results of R_h/R_l in Table 7.

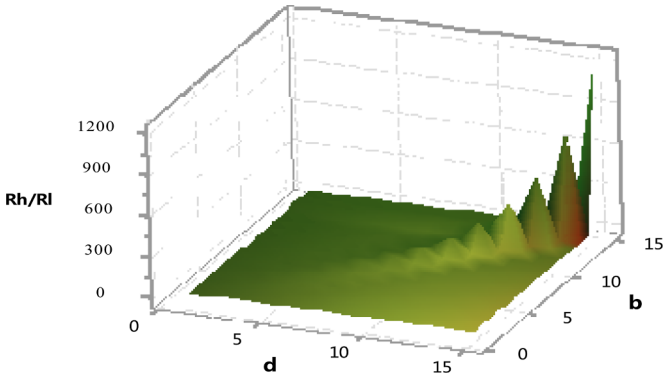


Fig. 4. 3D surface plot of b, d and R_h/R_l.

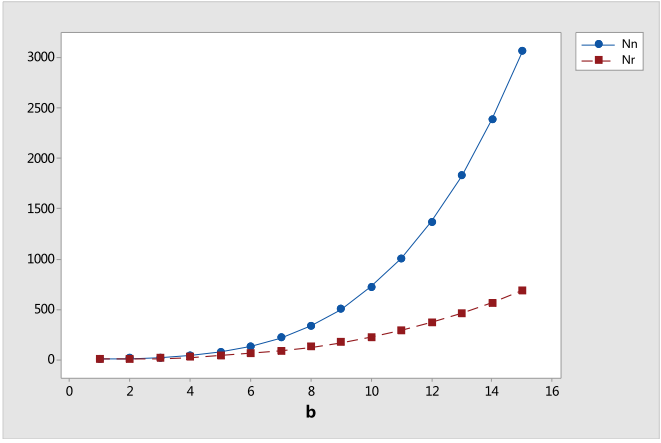


Fig. 5. Scatterplot of N_r, N_n and d, if $b = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$ for $d = 1$.

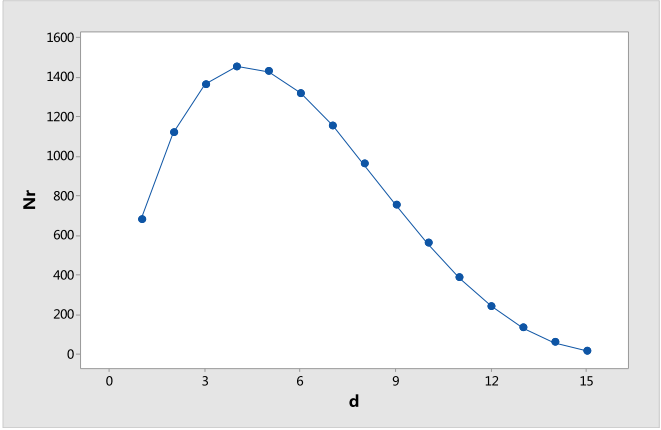


Fig. 6. Scatterplot of N_r and d, if $d = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$ for $b = 15$.

1 The larger b the higher N_n, N_r, R_n, R_r, R_l and R_h for fixed d owing to the factor that the occurrent probability is increasing with the value of state, e.g., the occurrent probabilities of states 10 and 11 are 0.05 and 0.93 of node 1, respectively. For example, Fig. 5 is the scatterplot of N_r, N_n and b, if $b = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$ for $d = 1$.

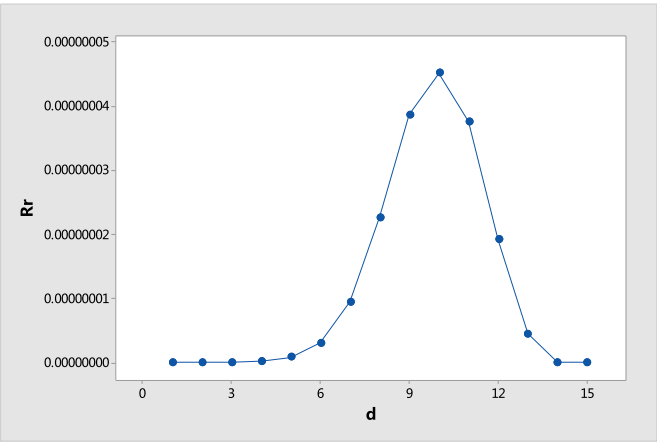


Fig. 7. Scatterplot of R_r and d, if $d = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$ for $b = 15$.

2 The values of N_n, R_n, R_l and R_h are decreased with the increasement of d, e.g., $N_n = 15, 5, 1$, $R_n = 0.000000004998411362 88$, $0.000000004933319821 60$, $0.000000004044607693 81$ if $d = 1, 2, 3$ for $b = 3$, respectively, in Tables 6 and 7. The above observation is owing to the larger d the lower chance to achieve the same.

3 The values of N_r and R_r are increased up to a peak then decreased with the increasement of d, e.g., $N_r = 84, 112, 105, 80, 50, 24, 7$, if $d = 1, 2, 3, 4, 5, 6, 7$ for $b = 7$, respectively, in Tables 6 and 7. For example, Fig. 6 is the scatterplot of N_r and d, if $d = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$ for $b = 15$, Fig. 7 is the scatterplot of R_r and d, if $d = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$ for $b = 15$.

4 The larger N_n and N_r, the higher values of R_n and R_r, respectively, and vice versa. The scatterplots of the final calculation results of N_r and R_r, N_n and R_n is shown in Fig. 8.

5 The larger b (but must not larger than the maximal capacity of each node), the greater number of N_n and N_r, respectively.

6 If b is larger than the maximal capacity of any node, both values of R_n and R_r are zero.

7 Interestingly, at least 99.999979% and 99.999799% of R_l and R_h are obtained from R_n from Tables 6 and 7, respectively, i.e., it appears as if the rework process is not very useful in increasing the network reliability under these assumptions provided in this paper.

Conclusions

Network reliability is a popular method for validating the designs and evaluating the performance of several practical systems. More than 25 previous studies have analyzed the rework network reliability problem [1,2]. However, these assumptions and limitations in the traditional rework network are not accurate and limit the applications of the rework network. Hence, a new general rework network is proposed making better assumptions without too many limitations along with a new algorithm based on three DFSs and a few simple properties of the problem to solve its reliability.

The experiments discussed in Section 5 show that the proposed algorithm not only helps us calculate the reliability of the general rework network, but it also aids in understanding the relationships among the values of b, d, N_n, N_r, R_n, R_r, R, δ_i, δ, and γ_i for all i. Hence, the proposed algorithm helps decision makers make better decisions regarding the rework process and can be extended for assessing other stochastic reliability problems.

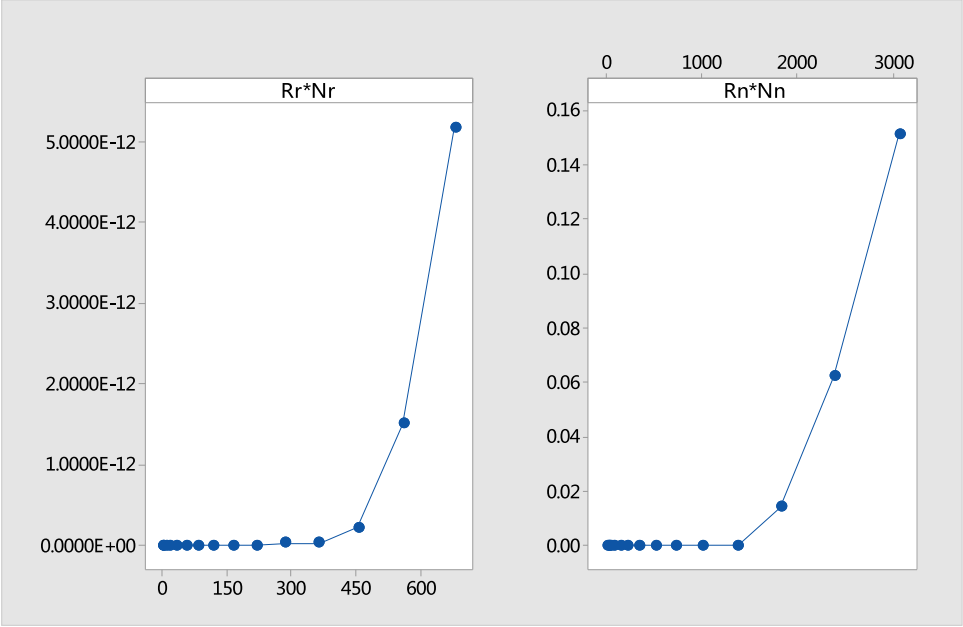


Fig. 8. Scatterplots of the final calculation results of N_r and R_r, N_n and R_n.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2020.107048.

Appendix A

Appendix A provides the proofs of Theorem 1.

Lemma 1 and lemma 2 discuss the relationship between p_x and q_x in Section 3.2.

Lemma 1. $p_{x,i} \le p_{x,j}$ and $q_{x,j} \le q_{x,i}$ if $j \le i$.

Proof. Based on the deterioration effect [8–10], the number of perfect products decreases with the number of processed nodes if no reworked process is conducted. Hence, we have $p_{x,i} \le p_{x,j}$ if $j \le i$. Similarly, the number of defects are increasing owing to the augmentation effect and we have $q_{x,j} \le q_{x,i}$ if $j \le i$.ÿ

Lemma 2.

$$p_{x,i} = b - \sum_{k=1}^{i} q_{x,k} = p_{x,i-1} - q_{x,i} \tag{9}$$

$$q_{x,i} = b - \sum_{k=1}^{i} p_{x,k} = p_{x,i-1} - p_{x,i}, \tag{10}$$

respectively.

Proof. The number of perfect products right after being processed in node 1 is the total number of products minus the number of defect products after being processed in node 1, i.e., $p_{x,1} = b - q_{x,1}$. Similarly, $p_{x,k} = b - \sum_{k=1}^{i} q_{x,k} = p_{x,i\text{-}1} - q_{x,i}$ and $q_{x,i} = b - \sum_{k=1}^{i} p_{x,k} = p_{x,i\text{-}1} - p_{x,i}$.

Lemma 3 discusses the relationship between π_y and θ_y in Section 3.2.

Lemma 3. 1. $\pi_{y,i} \le \pi_{y,j}$ and $\theta_{y\ x,j} \le \theta_{y,i}$ if $j \le i$.

2. $$\pi_{y,i} = b^* - \sum_{k=\beta}^{i} \theta_{y,k} = \pi_{y,i-1} - \theta_{y,i} \tag{11}$$

3. $$\theta_{y,i} = b^* - \sum_{k=\beta}^{i} \pi_{y,k} = \pi_{y,i-1} - \pi_{y,i}, \tag{12}$$

Proof. Analog to the proofs in Lemmas 1 and 2.

Lemma 4 considers a feasible solution is a vector without the rework process in Section 3.2.

Lemma 4. The solution $X = (p_x, \pi_y)$ is a feasible non-rework solution if $d \le p_{x,i} \le p_{x,j}$ for all $j \le i$ and π_y is a zero vector.

Proof. Directly follows from Lemma 1 and the definitions of feasible non-rework vector and X.

Lemma 5 considers a feasible solution formed by two vectors with the rework process in Section 3.2.

Lemma 5. The solution $X = (p_x, \pi_y)$ is a feasible non-rework solution if $p_{x,i} \le p_{x,j}$ for all $j \le i$, $\pi_{y,k} \le \pi_{y,l}$ for all $l \le k$, $p_{x,n} < d$, $d \le p_{x,\alpha}$, $\pi_{y,\alpha} \le q_{x,\alpha} = b - \sum_{k=1}^{\alpha} p_{x,k}$, and $d \le p_{x,n} + \pi_{y,n}$.

Proof. From Lemmas 1 and 3, we have $p_{x,i} \le p_{x,j}$ for all $j \le i$, $\pi_{y,a} \le \pi_{y,b}$ for all $b \le a$. From the proposed Assumption 1, a work process is necessary if $p_{x,n} < d$. All products outputted from node α are either perfect and sent to next node to process, defective and can be sent back to rework, or must be discarded. Hence, the solution X is required to have at least d units of non-defective products if $d > p_{x,\alpha}$, i.e., we must have $d \le p_{x,\alpha}$ and also $\pi_{y,\alpha} \le q_{x,\alpha} = b - \sum_{k=1}^{\alpha} p_{x,k}$ from Lemma 2. A feasible solution must have at least d units of non-defective products from both the first and the second processes in the end, i.e., $d \le p_{x,n} + \pi_{y,n}$.

Lemma 6. The proposed 3-DFS in Section 4.1 can identify all feasible solutions.

Lemma 7. Eq. (16) and Eq. (18) in the Case 2 of Section 4.2 are the occurrent probability of a feasible solution without and with requiring a rework process, respectively.

These complete the proofs of the Theorem 1.

Appendix B

Appendix B provides Tables 6 and 7 that shows the results of calculating the reliability using the proposed 3-DFS.

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